RESEARCH REVIEW

Smart Sensors, Connectivity, Optical Monitoring and AI for Solar Drying of Agricultural Produce

Purpose

A literature-based review of five recent research papers, with observations, methods, research gaps, proposed innovations, sensor connectivity, data-reading methods and a practical plan for a low-cost prototype.

Project direction

Design of a low-cost IoT and multimodal AI framework for solar drying, moisture estimation and colour-quality monitoring of agricultural produce.

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Important scope note: this is a focused review based on accessible publisher pages, abstracts and available article information. Some publisher full texts may require institutional access; the review does not claim that every full PDF was downloaded and read.

1. Executive summary

Recent work on smart solar drying is moving from simple temperature display toward connected systems that combine environmental sensing, mass-loss measurement, camera-based quality inspection, energy monitoring and AI-assisted decisions. The strongest opportunity for a student-scale project is to integrate these streams in a transparent, low-cost prototype and validate predictions against real reference measurements.

Main observation: temperature and relative humidity describe the drying environment, but they do not directly measure moisture inside a fruit or vegetable. Moisture estimation should therefore be calibrated against a reference method such as gravimetric moisture measurement. Likewise, colour estimates depend on lighting, camera settings and the crop, so image capture must be controlled.

Proposed contribution: synchronise air temperature/RH, tray mass change, energy use and standardised RGB images; create a labelled dataset; compare baseline models with multimodal models; and use a cautious, human-reviewed retraining workflow rather than an automatically self-changing model.

2. Five recent papers reviewed

Paper 1 — Progress in Smart Solar Drying Technologies and the Sustainability of Drying Systems: Systematic Review (2026)

Source: Nkolokosa, D., Loganathan, M., Waita, S. and Moses, J. (2026), *Journal of Food Process Engineering*, 49(3), e70439. DOI: https://doi.org/10.1111/jfpe.70439

Methods and focus: a systematic review of smart solar-drying technologies, including communication, remote control, automation, modelling and sustainability.

Observation: the literature increasingly treats drying as a monitored and controllable process rather than a passive cabinet. The review reports performance improvements across cited studies, but those values depend on the individual dryer, crop and experimental conditions and should not be treated as guaranteed results for a new prototype.

Connectivity / readings: this is a review rather than one hardware implementation. It supports considering sensor networks, remote monitoring and control, but does not prescribe one universal board or communication protocol.

Gap relevant to this project: performance metrics and sensor combinations vary widely, making direct comparison difficult. A small project should document calibration, crop, tray position, sampling interval and reference moisture method.

Paper 2 — Solar forced convection dryer for agriproducts monitored by IoT (2025)

Source: *Internet of Things*, 31 (2025), 101566. DOI: https://doi.org/10.1016/j.iot.2025.101566

Methods and focus: a forced-convection solar dryer with web-based remote supervision, real-time mass-loss measurement using a load cell, an embedded camera and control of drying-air mass flow.

Observation: combining mass change and images is more informative than relying on temperature alone. Mass loss can track drying progress, while images can reveal visible colour or appearance changes.

Connectivity / readings: sensor readings and camera observations are made available through an IoT/web monitoring approach. The exact sampling and communication details should be checked in the full paper before reproducing the design.

Gap relevant to this project: mass change is not automatically equal to water loss in every condition, and a camera cannot directly determine internal moisture. Both signals need reference validation and crop-specific testing.

Paper 3 — Analysis of electrical energy requirements from solar panels based on IoT on a Solar Dryer Dome in coffee drying (2026)

Source: Widianti, N., Amuddin and Joko Sumarsono (2026), *J-AGENT*, 4(1), 14–21. https://jurnal.unram.ac.id/index.php/agent/en/article/view/10961

Methods and focus: a seven-day monitoring setup using ESP32, PZEM-017 electrical measurement, a 16×2 I²C LCD, SD-card logging and ThingSpeak. Voltage, current, power and energy were logged at ten-minute intervals.

Reported observations: the paper reports setup-specific electrical values, including 10.82–14.39 V voltage, 2.52–3.17 A current, a 43.6 W peak power and 3155 Wh total energy over seven days. These figures describe that system only and are not design targets for another dryer.

Connectivity / readings: ESP32 can collect local sensor data, show values on an LCD, save records to an SD card and publish them to a cloud dashboard over Wi-Fi. This is a useful example of combining local and remote logging.

Gap relevant to this project: energy reporting should be linked to drying outcomes—such as water removed, drying time and colour retention—not reported in isolation. Power sensors must also be selected for the actual DC/AC circuit and current range.

Paper 4 — Predicting colour change in solar-dried fruits using neural networks: A multilayer perceptron approach

Source: publisher article page, *Journal of Stored Products Research* (ScienceDirect listing). https://www.sciencedirect.com/science/article/pii/S0022474X25002620

Methods and focus: a multilayer perceptron (MLP) predicts CIELAB colour outputs from nine inputs; genetic-algorithm architecture optimisation and Bayesian regularisation are described. The publisher summary reports a relatively small dataset (64 training/validation samples and 26 comparison samples).

Observation: machine learning can estimate colour changes, but small datasets may not represent different crops, cameras, lighting conditions or dryer designs. High reported performance on one dataset does not establish generalisation.

Reading method: camera images should be captured at a fixed distance and angle under controlled illumination. If possible, include a colour reference card and derive CIELAB L*, a*, b* and ΔE rather than using uncalibrated RGB values alone.

Gap relevant to this project: test on separate drying runs and, ideally, a separate batch. Avoid randomly splitting near-duplicate images from the same run across training and test sets, as that can inflate performance.

Paper 5 — A multimodal measurement and explainable AI framework for energy-efficient and quality-aware monitoring of food drying

Source: publisher article page, ScienceDirect. https://www.sciencedirect.com/science/article/pii/S0263224126024516

Methods and focus: the available publisher summary describes PLC-based sensors, photovoltaic energy monitoring, computer vision, multimodal data fusion and explainable AI. Image processing includes Mask R-CNN segmentation and features such as shrinkage, brightness, redness and CIE76 ΔE; a decision-tree-based classifier is also reported.

Observation: fusing process measurements and image features may support quality-aware decisions. Explainable methods can help users understand which measured factors influence a prediction.

Connectivity / readings: the reported framework combines several sensor streams and energy measurements. It is a more complex approach than a beginner prototype and may require PLC hardware and more compute than a low-cost ESP32-only design.

Gap relevant to this project: reported performance on a particular pepper dataset may not transfer to other produce. A smaller system should start with simpler features and models, then add deep-learning segmentation only when enough labelled images are available.

3. Cross-paper observations

- **Environmental sensing:** temperature and relative humidity are foundational for understanding drying conditions, but they are indirect indicators of product moisture.
- **Mass tracking:** load cells provide a useful drying-progress signal when trays are mechanically stable and readings are protected from fan vibration.
- **Optical sensing:** RGB cameras are relatively affordable for visible colour and defect monitoring. Repeatable lighting and calibration are essential.

- **Energy monitoring:** voltage, current, power and cumulative energy help evaluate the energy cost of drying, especially in solar-powered systems.
- **Connectivity:** ESP32 Wi-Fi, local SD-card storage and a cloud dashboard can form a practical low-cost architecture. Local logging should continue if Wi-Fi fails.
- **AI design:** begin with interpretable baselines; use more complex deep learning only when dataset size and compute resources justify it.
- **Validation:** report results by crop and drying run, compare estimates with reference measurements and clearly separate training, validation and test data.

4. Research gaps

G1. Reliable moisture ground truth Air RH and temperature do not directly reveal internal produce moisture. Predictions need comparison with a documented reference method, for example periodic gravimetric measurements.

G2. Low-cost multimodal integration Many systems focus on one signal or use industrial hardware. There is room to test a low-cost combination of environment, mass, image and energy data in one prototype.

G3. Tray-to-tray variation Airflow and temperature may differ by tray and location. Measurements should be associated with tray IDs and compared across positions.

G4. Repeatable colour measurement Lighting, shadows, camera exposure, white balance and distance can distort colour readings. A repeatable imaging enclosure or lighting protocol is needed.

G5. Generalisation across crops and batches Models trained on one fruit, vegetable or batch may fail on another. Evaluation should use separate runs and report crop-specific performance.

G6. Time alignment and missing data Camera images, load-cell readings and sensor logs may have different timestamps or missing records. Use a common clock, run ID and clear missing-data flags.

G7. Energy-to-quality trade-off Energy consumption is often reported separately from final moisture and colour quality. The system should examine energy per unit of water removed alongside product quality.

G8. Safe and stable automation AI predictions should not be the sole safety mechanism. Independent temperature limits, manual override and fail-safe fan behaviour are important.

G9. Calibration and reproducibility Sensor drift, placement and calibration influence results. Record sensor model, calibration checks, sampling interval and experimental conditions.

G10. Responsible self-learning Automatic model updates can degrade performance. New data should be labelled, checked, evaluated against a fixed test set and approved before deployment.

5. Proposed innovations for the student project

- **Innovation 1 — Synchronized data record:** store every measurement with timestamp, run ID, crop, tray ID and sensor status so that all data streams can be aligned.
- **Innovation 2 — Drying progress from multiple signals:** combine temperature/RH trends with tray mass loss and image-derived colour features rather than relying on a single sensor.
- **Innovation 3 — Affordable edge-plus-cloud design:** use ESP32 for acquisition and basic control; use a laptop or small edge computer for image processing and model inference; upload summaries to a dashboard when connected.
- **Innovation 4 — Explainable prediction:** show estimated drying stage, confidence or uncertainty, and the main inputs that contributed to the prediction.
- **Innovation 5 — Controlled model update:** collect new labelled runs, retrain offline, compare with the existing model and deploy only after performance and safety checks.
- **Innovation 6 — Quality-energy evaluation:** assess moisture endpoint, colour change, drying time and energy consumed together.

6. Proposed methodology

Phase A — Build and baseline

Build a small chamber or instrument an existing solar dryer. Keep electronics away from hot, humid air. Begin with temperature/RH monitoring and local data logging before adding camera and AI features.

Phase B — Sensor acquisition

Use an ESP32 development board. Select a temperature/RH sensor appropriate to the expected temperature and humidity range. Add a load cell with a suitable amplifier if mass tracking is required. Add a compatible voltage/current sensor only after documenting whether the monitored circuit is DC or AC and its voltage/current limits.

Phase C — Image acquisition

Mount the camera in a repeatable position. Capture images at a defined interval under fixed illumination, with minimal shadows and a stable background. Record the crop, tray and time for each image. Use a colour reference card during calibration where feasible.

Phase D — Ground-truth labels

At planned intervals, take a representative sample and determine reference moisture content using an approved laboratory or gravimetric procedure under supervisor guidance. Record mass before and after the reference procedure and document the calculation basis (wet basis or dry basis). Do not label sensor RH as product moisture.

Phase E — Data processing and modelling

Clean impossible readings, flag missing values, align timestamps, and create features such as temperature/RH history, mass-loss rate, colour values and energy use. Start with linear regression, random forest or gradient-boosted trees for moisture estimation; use regression for colour values and simple classifiers for drying stage. Consider CNN/segmentation models only when the image dataset is sufficient.

Phase F — Testing

Use separate complete drying runs for training, validation and testing. Compare predictions with reference labels using MAE and RMSE for moisture/colour regression, and accuracy, precision, recall and F1 for classification. Report errors by crop and tray position, not only one overall score.

7. Connectivity architecture

Layer	Suggested option	Purpose / notes
Sensors	Temperature/RH; optional load cell; power sensor; RGB camera	Measure environment, mass trend, energy and visible appearance. Camera may connect directly to the edge computer rather than ESP32.
Microcontroller	ESP32 DevKit	Reads supported digital/analogue sensors, timestamps data, performs simple rules and sends records over Wi-Fi.
Local storage	MicroSD or laptop log	Preserves data during network outages; use CSV/JSON with stable column names.
Network	Wi-Fi; MQTT or HTTPS	MQTT suits streaming telemetry; HTTPS/REST suits periodic uploads. Use authentication and avoid exposing the device directly to the public internet.
Edge processing	Laptop / small computer	Processes images, runs models and creates predictions. This reduces pressure on the microcontroller.
Dashboard	ThingSpeak or a simple web dashboard	Plots temperature/RH, mass, power, image-derived colour and predicted drying stage. Cloud is optional for local operation.
Control	MOSFET/relay driver and low-voltage DC fan	Never power a fan directly from an ESP32 GPIO. Use a correctly rated driver, separate supply as needed, manual override and independent temperature limits.

8. Reading methods and recommended sampling

Sampling intervals below are practical starting points for a student prototype, not universal values reported by all five papers. Adjust them after checking sensor response, expected drying speed and data-storage needs.

Signal	Suggested starting interval	Reading / handling method
Temperature and RH	Every 10–60 seconds	Read digitally, check range and sensor status, log timestamp and sensor location.
Tray mass	Every 30–60 seconds after stabilisation	Tare before the run; average stable samples; isolate load cell from vibration and airflow forces.
Power / energy	Every 10–60 seconds; store cumulative energy	Use a rated meter for the actual circuit; record voltage, current, power and cumulative Wh.
RGB image	Every 5–15 minutes, or at planned checkpoints	Fixed lighting, distance and exposure; save original image plus run/tray/time metadata.
Reference moisture	At planned checkpoints / end points	Use a documented reference procedure and representative samples; record basis and units.
Model prediction	After each valid data update or image	Log model version, prediction, confidence/uncertainty and whether the result was reviewed.

9. Evaluation plan

- **Moisture estimation:** MAE and RMSE against the reference moisture values; report units and whether values use wet or dry basis.
- **Colour prediction:** error in L*, a*, b* and ΔE, using a consistent colour space and capture protocol.
- **Defect detection / segmentation:** precision, recall, F1 and intersection-over-union if pixel-level labels are created.
- **Drying-stage classification:** accuracy, macro-F1 and a confusion matrix; check whether classes are imbalanced.
- **Energy and process:** total energy, drying duration, mass lost and final quality measurements. Compare runs under documented conditions.
- **Robustness:** evaluate different runs, tray positions and, if in scope, a second crop. State limitations honestly.

10. Research questions

- Can temperature/RH plus mass-loss history estimate product moisture more accurately than temperature/RH alone?
- Does adding calibrated colour features improve drying-stage or quality prediction?
- How much variation exists between tray positions, and can sensor placement reveal it?
- What is the trade-off between energy consumption, drying duration and colour retention?
- Can a low-cost ESP32 plus local edge computer provide reliable monitoring when Wi-Fi is intermittent?

11. Suggested research title and abstract

Title: Design of a Low-Cost IoT and Multimodal AI Framework for Solar Drying, Moisture Estimation and Colour-Quality Monitoring of Agricultural Produce

Draft abstract (proposal stage): This project proposes a low-cost monitoring framework for solar drying of agricultural produce using IoT sensors and multimodal data analysis. An ESP32-based node will collect drying-air temperature and relative humidity, with optional tray mass and electrical-energy measurements. A camera under controlled illumination will record visible colour changes. Sensor records and images will be time-aligned and labelled using reference moisture measurements and documented drying runs. Baseline machine-learning models will be compared with multimodal models to estimate moisture or drying stage and to track colour change. The system will combine local data logging with Wi-Fi-based monitoring and retain safe manual control of the dryer. Evaluation will use held-out drying runs and metrics appropriate to each prediction task. The proposed contribution is an affordable, reproducible integration of process, quality and energy data; performance claims will be made only after experimental validation.

Status: proposed method only. Do not present the abstract as completed experimental results until hardware, data collection and testing have been performed.

12. References and access links

• 1. Nkolokosa et al. (2026). Progress in Smart Solar Drying Technologies and the Sustainability of Drying Systems: Systematic Review. Journal of Food Process Engineering, 49(3), e70439.
https://doi.org/10.1111/jfpe.70439

• 2. Solar forced convection dryer for agriproducts monitored by IoT. Internet of Things, 31 (2025), 101566.
https://doi.org/10.1016/j.iot.2025.101566

• 3. Widianti, Amuddin & Sumarsono (2026). Analysis of electrical energy requirements from solar panels based on IoT on a Solar Dryer Dome in coffee drying. J-AGENT, 4(1), 14–21.
https://jurnal.unram.ac.id/index.php/agent/en/article/view/10961

• 4. Predicting color change in solar-dried fruits using neural networks: A multilayer perceptron approach. Publisher article page.
https://www.sciencedirect.com/science/article/pii/S0022474X25002620

• 5. A multimodal measurement and explainable AI framework for energy-efficient and quality-aware monitoring of food drying. Publisher article page.
https://www.sciencedirect.com/science/article/pii/S0263224126024516

• Additional useful reading: moisture detection in mango and papaya slices using a multimode-interference optical sensor.
https://pubmed.ncbi.nlm.nih.gov/41305110/

• Additional review: hyperspectral imaging for rapid assessment of moisture and chromaticity during food drying.
https://www.sciencedirect.com/science/article/abs/pii/S0168169924004927

• Additional review: Smart drying of agricultural produce: an Industry 4.0 perspective.
https://doi.org/10.1016/j.tifs.2026.105557

13. Final conclusion

The reviewed literature supports a project that combines sensor monitoring, IoT connectivity, mass tracking, camera-based colour analysis and AI. The most defensible research gap is not simply to add more sensors; it is to synchronise and validate low-cost multimodal measurements, quantify uncertainty, test across independent drying runs and connect product quality to energy use. A staged build—environmental sensing first, then mass and energy, then controlled imaging, then AI—is the most practical route for a reliable student prototype.